

1. Building Docker image

Make sure that you are having application code and Dockerfile in the same folder

```
ubuntu@ip-172-31-92-238:~$ sudo su -
root@ip-172-31-92-238:~# cd /root/app/
root@ip-172-31-92-238:~/app# ls -l
total 8
-rw-r--r-- 1 root root 302 Mar 24 02:52 app.py
-rw-r--r-- 1 root root  6 Mar 24 02:52 requirements.txt
root@ip-172-31-92-238:~/app# cat app.py
from flask import Flask, jsonify

app = Flask(__name__)

@app.route("/")
def home():
    return jsonify({"message": "Hello All Welcome to Dvs GenAI Sessions!"})

@app.route("/health")
def health():
    return jsonify({"status": "OK"})

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
root@ip-172-31-92-238:~/app# cat requirements.txt
flask
root@ip-172-31-92-238:~/app#
```

```
root@ip-172-31-92-238:~/app# cat Dockerfile
FROM python:3.11
WORKDIR /opt/app
COPY app.py /opt/app/app.py
COPY requirements.txt /opt/app/
EXPOSE 5000
RUN pip install -r requirements.txt
CMD ["python", "app.py"]
```

root@ip-172-31-92-238:~/app# cat Dockerfile

FROM python:3.11

WORKDIR /opt/app

COPY app.py /opt/app/app.py

COPY requirements.txt /opt/app/

EXPOSE 5000

RUN pip install -r requirements.txt

CMD ["python", "app.py"]

If you want to create the image then you can execute your below

docker build -t "dvsgenaiapp:v1.0" .

Note: you can give any name

```
root@ip-172-31-92-238:~/app# docker build -t "dvsgenaiapp:v1.0" .
[+] Building 1.5s (8/9)
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 206B
=> [internal] load metadata for docker.io/library/python:3.11
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.11@sha256:4651409ff55024245f70920259ec
=> => resolve docker.io/library/python:3.11@sha256:4651409ff55024245f70920259ec
=> [internal] load build context
=> => transferring context: 388B
=> [2/5] WORKDIR /opt/app
=> [3/5] COPY app.py /opt/app/app.py
=> [4/5] COPY requirements.txt /opt/app/
=> [5/5] RUN pip install -r requirements.txt
```

Verify images

```
=> => unpacking to docker.io/library/dvsgenaiapp:v1.0
root@ip-172-31-92-238:~/app# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
centos:8	a27fd8080b51	340MB	83.5MB	
dvsgenaiapp:v1.0	a61dc47d69f5	1.62GB	415MB	
hello-world:latest	6ae972dlac53	25.9kB	9.49kB	U
python:3.11	4651409ff550	1.61GB	427MB	U

```
root@ip-172-31-92-238:~/app#
```

2.Let's create a container

docker run --name myapp -d -p 5000:5000 dvsgenaiapp:v1.0

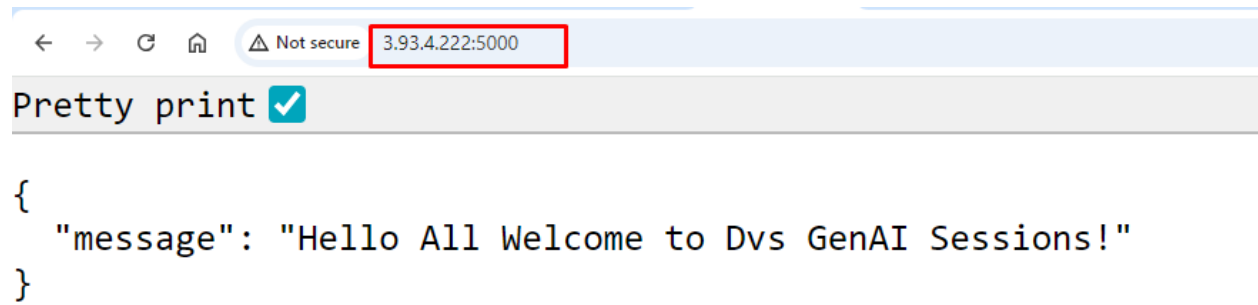
docker ps

```
root@ip-172-31-92-238:~/app# docker run --name myapp -d -p 5000:5000 dvsgenaiapp:v1.0
558f1888a2959539930bfd9d6931c4d738b0c2d084f6f0f97d7df32b41627964
root@ip-172-31-92-238:~/app# docker ps
```

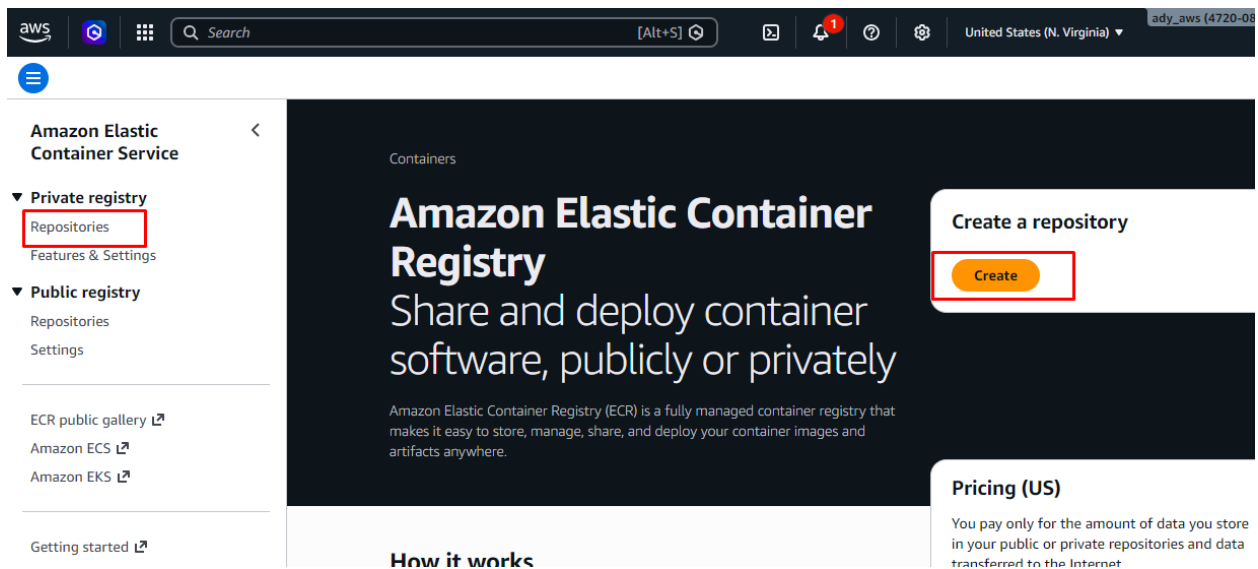
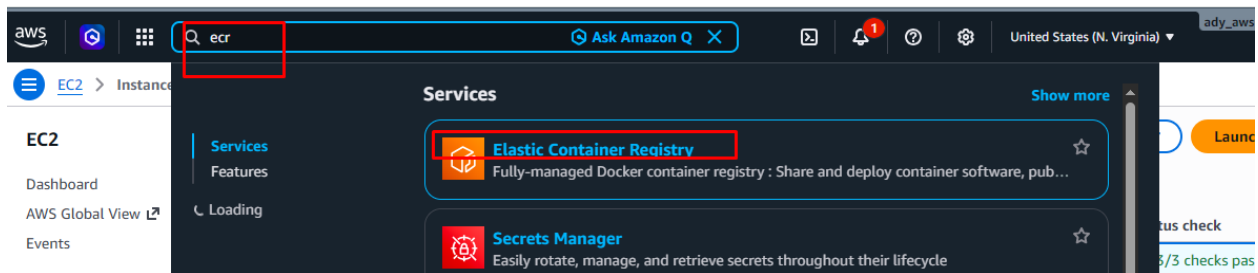
CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	P
558f1888a295	dvsgenaiapp:v1.0	"python app.py"	5 seconds ago	Up 5 seconds	0
6e1a13e2ca21	python:3.11	"/bin/bash"	24 hours ago	Up 24 hours	

```
root@ip-172-31-92-238:~/app#
```

Note: *Make sure you are opening port 5000 in the security group or open all ports in the server security group*



3.Working with Registry



Create private repository

General settings

Repository name

Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.

472008046327.dkr.ecr.us-east-1.amazonaws.com/ genaiapp/app1

13 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters _-./.

Image tag settings [Info](#)

Image tag mutability

Choose the tag mutability setting.

☒ **Mutable**
Image tags can be overwritten.

☐ **Immutable**
Image tags can't be overwritten.

Mutable tag exclusions

Tags that match these filters will be immutable (can't be overwritten). Using wildcards (*) will match zero or more image tag characters.

[Add filter](#)

Filters must only contain letters, numbers, and special characters (_ - . / *). Each filter is limited to 128 characters, 2 wildcards (*), and you can add up to 5 filters in the exclusion list.

Encryption settings [Info](#)

 The encryption settings for a repository can't be changed once the repository is created.

Encryption configuration

By default, repositories use the industry standard Advanced Encryption Standard (AES) encryption. You can optionally choose to use a key stored in the AWS Key Management Service (KMS) to encrypt the images in your repository.

☒ **AES-256**
Industry standard Advanced Encryption Standard (AES) encryption

☐ **AWS KMS**
AWS Key Management Service (KMS)

► Image scanning settings - *deprecated*

[Cancel](#)

[Create](#)

> Repositories

✔ Successfully created private repository, genaiapp/app1

Private repositories (1)



View push commands

Delete

Actions ▾

Create new

🔍 Search by repository substring

Repository name	URI	Created at	Tag immutability	Encryption type
☐ genaiapp/app1	472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1	March 25, 2026, 06:33:48 (UTC+04)	Mutable	AES-256

4.Pushing images to the repository

Amazon ECR > Private registry > Repositories > genaiapp/app1

Amazon Elastic Container Service

- ▼ Private registry
 - Repositories
 - [genaiapp/app1](#)
 - Features & Settings
- ▼ Public registry
 - Repositories
 - Settings

ECR public gallery [↗](#)
Amazon ECS [↗](#)
Amazon EKS [↗](#)

genaiapp/app1

Summary | **Images** | Lifecycle policy | Permissions | Repository tags

Images Info Delete Copy URI Details Scan **View push commands**

🔍 Filter active images

☐	Image tags ▾	Type	Created at ▾	Image size (MB) ▾	Image digest	Last pulled at ▾
No active images						

Push commands for genaiapp/app1



macOS / Linux

Windows

Make sure that you have the latest version of the AWS CLI and Docker installed. For more information, see [Getting Started with Amazon ECR](#).

Use the following steps to authenticate and push an image to your repository. For additional registry authentication methods, including the Amazon ECR credential helper, see [Registry Authentication](#).

1. Retrieve an authentication token and authenticate your Docker client to your registry. Use the AWS CLI:

```
aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 472008046327.dkr.ecr.us-east-1.amazonaws.com
```

Note: If you receive an error using the AWS CLI, make sure that you have the latest version of the AWS CLI and Docker installed.

2. Build your Docker image using the following command. For information on building a Docker file from scratch see the instructions [here](#). You can skip this step if your image is already built:

```
docker build -t genaiapp/app1 .
```

3. After the build completes, tag your image so you can push the image to this repository:

```
docker tag genaiapp/app1:latest 472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:latest
```

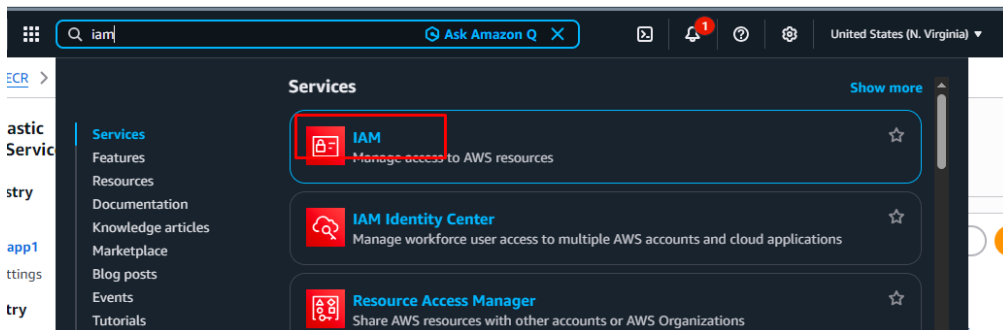
4. Run the following command to push this image to your newly created AWS repository:

```
docker push 472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:latest
```

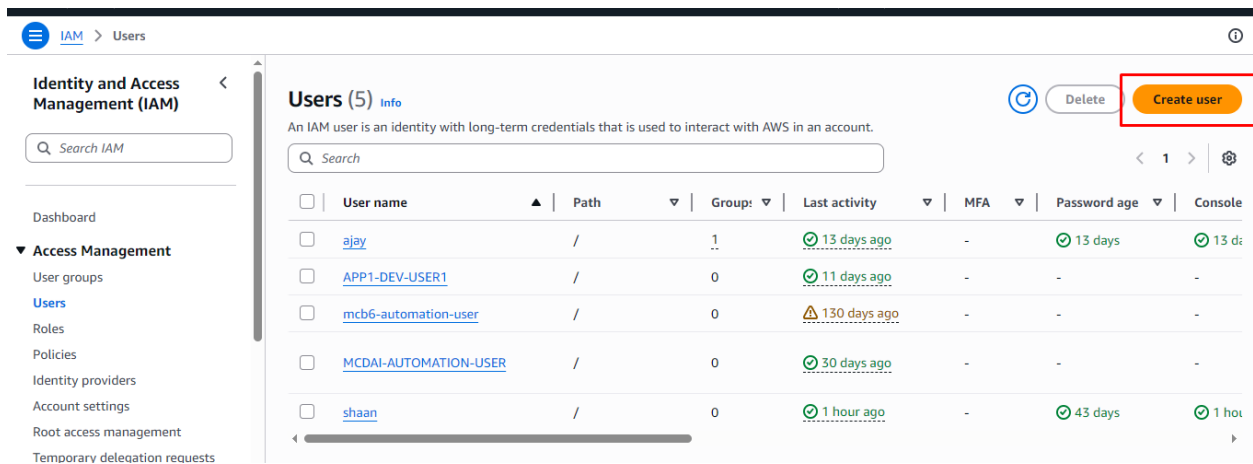
Close

5. Generate credentials to push images to ECR

IAM

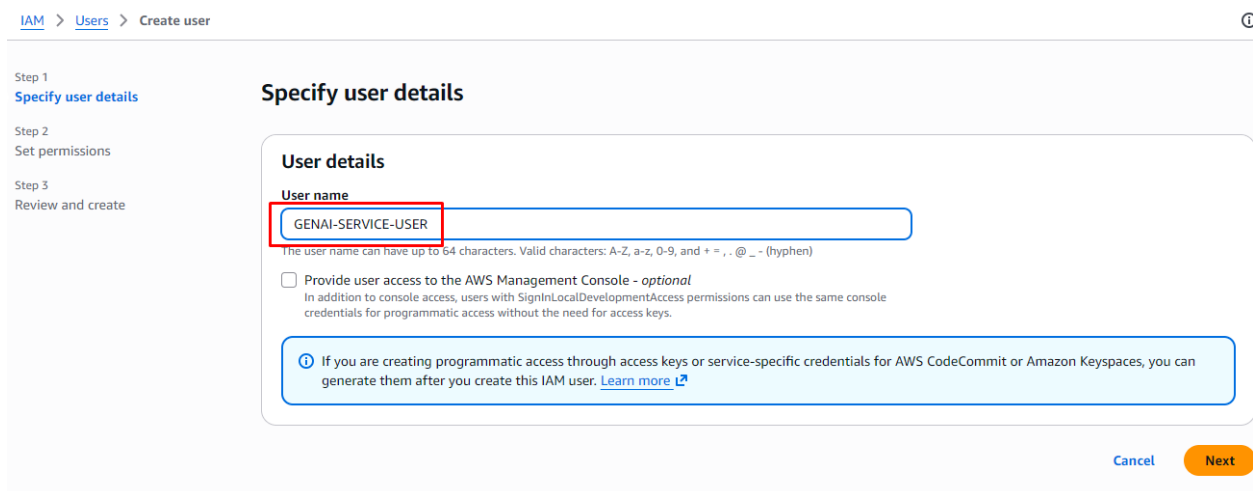


Create a service account and generate the credentials



The screenshot shows the AWS IAM console 'Users' page. On the left is a navigation menu with 'Identity and Access Management (IAM)' and 'Access Management' sections. The main area is titled 'Users (5)' and includes a search bar and a table of existing users. The 'Create user' button in the top right corner is highlighted with a red box.

☐	User name	Path	Group	Last activity	MFA	Password age	Console
☐	ajay	/	1	13 days ago	-	13 days	13 d
☐	APP1-DEV-USER1	/	0	11 days ago	-	-	-
☐	mcb6-automation-user	/	0	130 days ago	-	-	-
☐	MCDAI-AUTOMATION-USER	/	0	30 days ago	-	-	-
☐	shaan	/	0	1 hour ago	-	43 days	1 hou



The screenshot shows the 'Specify user details' step of the 'Create user' process. The 'User name' field is highlighted with a red box and contains the text 'GENAI-SERVICE-USER'. Below the field is a note about valid characters. There is an optional checkbox for 'Provide user access to the AWS Management Console'. At the bottom right are 'Cancel' and 'Next' buttons.

Specify user details

User details

User name

GENAI-SERVICE-USER

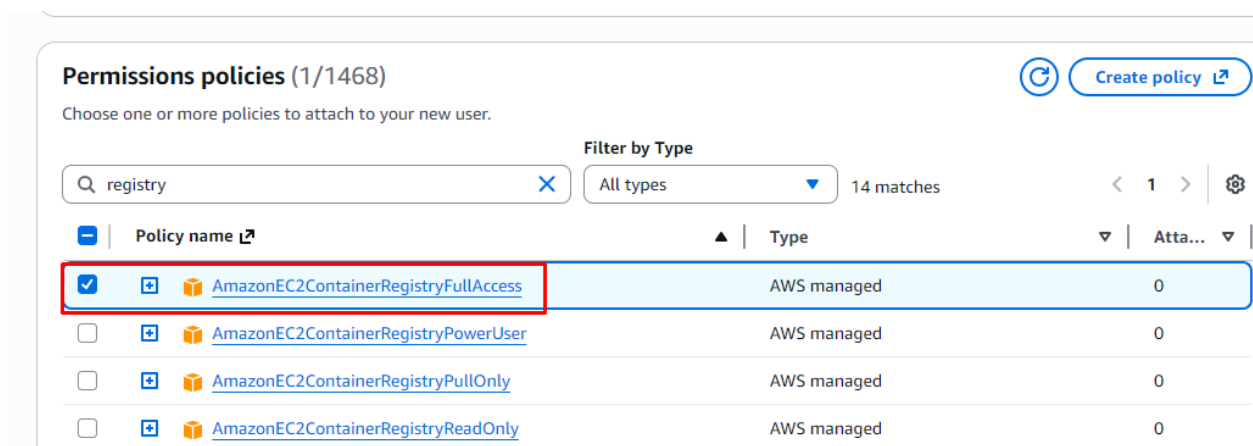
The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☐ Provide user access to the AWS Management Console - optional

In addition to console access, users with SignInLocalDevelopmentAccess permissions can use the same console credentials for programmatic access without the need for access keys.

If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel Next



The screenshot shows the 'Permissions policies' step. It displays a list of policies filtered by 'registry'. The 'AmazonEC2ContainerRegistryFullAccess' policy is highlighted with a red box. The table lists policy names, types, and attachment counts.

Permissions policies (1/1468)

Choose one or more policies to attach to your new user.

Filter by Type

registry 14 matches

☐	Policy name	Type	Atta...
☒	AmazonEC2ContainerRegistryFullAccess	AWS managed	0
☐	AmazonEC2ContainerRegistryPowerUser	AWS managed	0
☐	AmazonEC2ContainerRegistryPullOnly	AWS managed	0
☐	AmazonEC2ContainerRegistryReadOnly	AWS managed	0

☐		AWSManagedReadOnlyAccess	AWS managed 0
☐		AWSServiceCatalogAppRegistryReadOnlyAccess	AWS managed 0
☐		AWSServiceCatalogAppRegistryServiceRolePolicy	AWS managed 0
☐		ROSAImageRegistryOperatorPolicy	AWS managed 0

► Set permissions boundary - optional

Cancel Previous **Next**

Permissions summary

Name ↗	Type	Used as
AmazonEC2ContainerRegistryFullAccess	AWS managed	Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

Cancel Previous **Create user**

6.Generate access keys

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access Management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

[View user](#)

ARN	Console access	Access key ID
arn:aws:iam::472008046327:user/GENAI-SERVICE-USER	Disabled	Create access key
Created	Last console sign-in	
March 25, 2026, 06:37 (UTC+04:00)	-	

Permissions Groups Tags **Security credentials** Last Accessed

Permissions policies (1)

Permissions are defined by policies attached to the user directly or through groups.

[Remove](#) [Add permissions](#)

Filter by Type

Search All types

< 1 >

Set description tag - optional [Info](#)

The description for this access key will be attached to this user as a tag and shown alongside the access key.

Description tag value

Describe the purpose of this access key and where it will be used. A good description will help you rotate this access key confidently later.

Maximum 256 characters. Allowed characters are letters, numbers, spaces representable in UTF-8, and: _ . : / = + - @

[Cancel](#)

[Previous](#)

[Create access key](#)

- Step 1
Access key best practices & alternatives
- Step 2 - optional
Set description tag
- Step 3
Retrieve access keys

Retrieve access keys [Info](#)

Access key

If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key

Secret access key

 AKIAW3ZO7G33RPTXUHWY

 ***** [Show](#)

Access key best practices

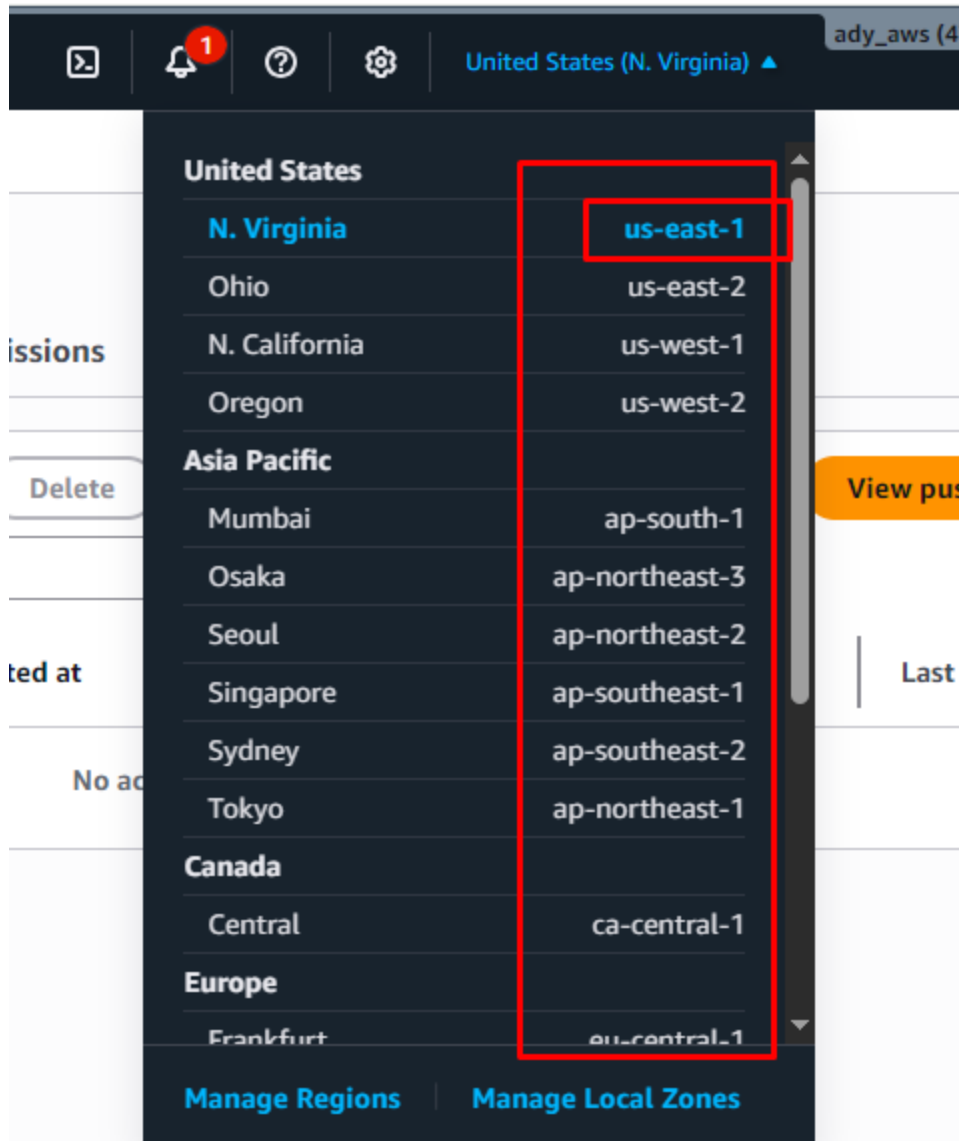
- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

7. Configure the keys in the server/laptop

Installing aws-cli software

```
snap install aws-cli --classic
```

```
root@ip-172-31-92-238:~/app# snap install aws-cli --classic
error: unknown flag `classic'
root@ip-172-31-92-238:~/app# snap install aws-cli --classic
Download snap "aws-cli" (2052) from channel "v2/stable" 100% 9
```



aws configure

```
root@ip-172-31-92-238:~/app# snap install aws-cli --classic
aws-cli (v2/stable) 2.34.16 from Amazon Web Services (aws✓) installed
root@ip-172-31-92-238:~/app# aws configure
AWS Access Key ID [None]: AKIAW3ZO7G33RFP17GMY
AWS Secret Access Key [None]: QRWjpWcsRupYuDxGmpYVgxkoIM2n6G3a10J
Default region name [None]: us-east-1
Default output format [None]: json
root@ip-172-31-92-238:~/app#
```

8. Finally let's push our images to our repository

```
aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 472008046327.dkr.ecr.us-east-1.amazonaws.com
```

```
Default output format [None]: json
root@ip-172-31-92-238:~/app# aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 472008046327.dkr.ecr.us-east-1.amazonaws.com

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
root@ip-172-31-92-238:~/app#
```

```
root@ip-172-31-92-238:~/app# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
centos:8	a27fd8080b51	340MB	83.5MB	
dvsgenaiapp:v1.0	a61dc47d69f5	1.62GB	415MB	U
hello-world:latest	6ae972d1ac53	25.9kB	9.49kB	U
python:3.11	4651409ff550	1.61GB	427MB	U

```
root@ip-172-31-92-238:~/app#
```

You have to rename the image as per your repository standards

```
docker tag dvsgenaiapp:v1.0 472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0
```

```
python:3.11 4651409ff550 1.61GB 427MB U
root@ip-172-31-92-238:~/app# docker tag dvsgenaiapp:v1.0 472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0
root@ip-172-31-92-238:~/app# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
python:3.11	4651409ff550	1.61GB	427MB	U

```

c-1.amazonaws.com/genaiapp/app1:v1.0
root@ip-172-31-92-238:~/app# docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	In Use
472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0	a61dc47d69f5	1.62GB	415MB	U
centos:8	a27fd8080b51	340MB	83.5MB	U
dvsgenaiapp:v1.0	a61dc47d69f5	1.62GB	415MB	U
hello-world:latest	6ae972d1ac53	25.9kB	9.49kB	U
python:3.11	4651409ff550	1.61GB	427MB	U

```

root@ip-172-31-92-238:~/app#

```

Push image to repository

docker push 472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0

```

python:3.11 4651409ff550 1.61GB 427MB U
root@ip-172-31-92-238:~/app# docker push 472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0
The push refers to repository [472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1]
d85ffcdc441e: Pushed
6f903cac4871: Pushed
8f6ad858d0a4: Pushed
8cd65a420aac: Pushed
8688d0f2f567: Pushed
454b2175e029: Pushed
2686c1d19a2c: Pushed
dcd193def241: Pushed
b012eb15dff0: Pushed
ee3a0e7d77f0: Pushed
66063df90a44: Pushed
1589b6e505d3: Pushed
v1.0: digest: sha256:a61dc47d69f540aa366a0706ceb950705da8eee0ed0720607d997fc8ed07ac5
8 size: 856

```

Amazon ECR > Private registry > Repositories > genaiaapp/app1

Amazon Elastic Container Service

Private registry

Repositories

genaiaapp/app1

Attributes & Settings

Public registry

Repositories

Settings

Public gallery

Amazon ECS

Amazon EKS

genaiaapp/app1

Summary Images Lifecycle policy Permissions Repository tags

Images (3) Info

Filter active images

☐	Image tags	Type	Created at	Image size (MB)	Image digest	Last pulled at
☐	v1.0	Image Index	March 25, 2026, 06:45:01 (UTC+04)	415.38	sha256:a61...	-
☐	-	Image	March 25, 2026, 06:45:01 (UTC+04)	415.38	sha256:f3f5...	-

9. Final Verification

Deleting all containers

```
root@ip-172-31-92-238:~/app# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
PORTS         NAMES
f6676c23ae02   dvsgenaiapp:v1.0 "python app.py"         22 minutes ago Up 22 minutes
0.0.0.0:5001->5000/tcp, [::]:5001->5000/tcp myapp1
558f1888a295   dvsgenaiapp:v1.0 "python app.py"         27 minutes ago Up 27 minutes
0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp myapp
6e1a13e2ca21   python:3.11    "/bin/bash"             24 hours ago   Up 24 hours
myc1

root@ip-172-31-92-238:~/app# docker rm -f myapp1 myapp myc1
myapp1
myapp
myc1
root@ip-172-31-92-238:~/app#
```

```
root@ip-172-31-92-238:~/app# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS         NAMES
root@ip-172-31-92-238:~/app#
```

Delete all images

```
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
root@ip-172-31-92-238:~/app# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0	a61dc47d69f5	1.62GB	415MB	U
centos:8	a27fd8080b51	340MB	83.5MB	U
dvsgenaiapp:v1.0	a61dc47d69f5	1.62GB	415MB	U
hello-world:latest	6ae972d1ac53	25.9kB	9.49kB	U
python:3.11	4651409ff550	1.61GB	427MB	

```
root@ip-172-31-92-238:~/app#
```

`docker rmi -f a61dc47d69f5 a27fd8080b51 a61dc47d69f5 6ae972d1ac53 4651409ff550`

```
python:3.11 4651409ff550 1.61GB 427MB
root@ip-172-31-92-238:~/app# docker rmi -f a61dc47d69f5 a27fd8080b51 a61dc47d69f5 6ae972d1ac53 4651409ff550
Untagged: 472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0
Untagged: dvsgenaiapp:v1.0
Deleted: sha256:a61dc47d69f540aa366a0706ceb950705da8eee0ed0720607d997fc8ed07ac58
Untagged: centos:8
Deleted: sha256:a27fd8080b517143cbbbab9dfb7c8571c40d67d534bbdee55bd6c473f432b177
Untagged: hello-world:latest
Deleted: sha256:6ae972d1ac530d897fd03238e11e16157498834b49574f21a0c7d0f51b58f84d
Untagged: python:3.11
Deleted: sha256:4651409ff55024245f70920259ecee27f2109b0dc0e502a10bce541f1072cd66
Error response from daemon: No such image: a61dc47d69f5:latest
root@ip-172-31-92-238:~/app# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
-------	----	------------	--------------	-------

```
root@ip-172-31-92-238:~/app#
```

Amazon ECR

>

Private registry

>

Repositories

>

genaiapp/app1

Amazon Elastic Container Service

<

▼ Private registry

Repositories

genaiapp/app1

Features & Settings

▼ Public registry

Repositories

Settings

ECR public gallery

Amazon ECS

Amazon EKS

genaiapp/app1

Summary | Images | Lifecycle policy | Permissions | Repository tags

Repository details

Repository name

genaiapp/app1

Created at

March 25, 2026, 06:33:48 (UTC+04)

Repository ARN

arn:aws:ecr:us-east-1:472008046327:repository/genaiapp1

Repository URI

472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1

Tag mutability

Mutable

Tag mutability exclusions

-

Encryption type

AES-256

Scan frequency

Manual

docker pull 472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0

```
root@ip-172-31-92-238:~/app# docker pull 472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0
v1.0: Pulling from genaiapp/app1
dcd193def241: Download complete
Digest: sha256:a61dc47d69f540aa366a0706ceb950705da8eee0ed0720607d997fc8ed07ac58
Status: Downloaded newer image for 472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0
472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0
root@ip-172-31-92-238:~/app# docker images
```

IMAGE	ID	DISK USAGE	Info →	U	In Use
472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0	a61dc47d69f5	1.62GB			

```
root@ip-172-31-92-238:~/app#
```

docker run -d --name myapp -p 80:5000 472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0

```
d88d45d62eed0b00a289939e08cb328eac0c30116ca11bd4cd8bcc019ebd414
root@ip-172-31-92-238:~/app# docker ps
```

CONTAINER ID	IMAGE	STATUS	PORTS	COM
d88d45d62eed	472008046327.dkr.ecr.us-east-1.amazonaws.com/genaiapp/app1:v1.0	Up 1 second	0.0.0.0:80->5000/tcp, [::]:80->5000/tcp	py

```
python app.py"
myapp
root@ip-172-31-92-238:~/app#
```

http://3.93.4.222:80

← → ↻ 🏠 ⚠ Not secure 3.93.4.222

Pretty print ✓

```
{
  "message": "Hello All Welcome to Dvs GenAI Sessions!"
}
```